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| **Status** | Finished |
| **Started** | Monday, 26 October 2026, 6:52 PM |
| **Completed** | Monday, 26 October 2026, 6:56 PM |
| **Duration** | 3 mins 35 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(n log n)
b. O(1)
c. O(log n)
d. O👎 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

What is the best-case time complexity of linear search?

a. O(log n)
b. O👎
c. O(1) ✓
d. O(n log n)

Question **3**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search backtracks to find duplicate elements
b. The search stops immediately ✓
c. The search continues until the end of the list
d. The search starts over from the beginning

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There must be a mechanism to delete and/or insert elements in the list ✓

c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **5**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. When the item is not the array at all

b. Item is the last element in the array or item is not there at all

c. When the item is the last element in the array

d. When the item is somewhere in the middle of the array ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Jump search

b. Binary search

c. Linear search ✓

d. Hashing

Question **7**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

a. The search ends successfully ✓

b. The search continues in the left sublist

c. The search continues in the right sublist

d. The list is sorted

Question **8**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list
b. In the left sublist
c. At the beginning of the list
d. In the right sublist ✓

Question **9**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

a. Merge search
b. Bubble search
c. Quick search
d. Linear search ✓

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

a. In the left sublist ✓
b. At the beginning of the list
c. In the right sublist
d. In the entire list

Question **11**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Determining whether an element is present in a list ✓
b. Sorting elements in a list
c. Deleting elements from a list
d. Inserting elements into a list

Question **12**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. n/2
b. log n
c. n ✓
d. 1

Question **13**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is large and unsorted
c. When the list is small and sorted
d. When the list is small and unsorted ✓

Question **14**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

a. Linear search ✓
b. Hashing
c. Binary search
d. Dynamic search

Question **15**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. n ✓
c. log n
d. n/2